

Homework Assignment: Medical Image Generation & Translation with CycleGAN and Diffusion Models

Due Date: 2025/04/12

In this assignment, you will:

1. Compare CycleGAN and Diffusion Models for medical image generation and style transfer.
2. Explore architectural and training details of CycleGAN for MRI T1 $\leftrightarrow$ T2 translation.
3. Experiment with diffusion-based models (DDPMs) for a similar image-to-image translation task.
4. Evaluate the models qualitatively (visual inspection) and quantitatively (SSIM, PSNR, etc.).

Part A: Conceptual Questions

Provide detailed written answers.

1. GANs vs. Diffusion Models

1. Comparison of Objective Functions
 - Explain how the adversarial training objective of CycleGAN differs from the denoising objective in a diffusion model.
 - Why might a diffusion-based model be more stable during training compared to a GAN?
2. Role of Cycle Consistency
 - Summarize why CycleGAN needs cycle-consistency losses ($X \rightarrow Y \rightarrow X$ and $Y \rightarrow X \rightarrow Y$).
 - In your opinion, is a similar “cycle-consistency” concept relevant to diffusion-based approaches for image translation? Why or why not?
3. Sampling and Mode Collapse
 - In GANs, mode collapse is a common pitfall. How do diffusion models mitigate or avoid mode collapse, and what trade-offs might arise (e.g., sampling speed)?

2. CycleGAN Architecture & Training

1. U-Net

- Under what circumstances might a U-Net style generator be preferred for medical image translation tasks?

2. PatchGAN Discriminator

- Explain the concept of a PatchGAN discriminator.
- What benefits does a PatchGAN provide when dealing with high-resolution medical images?

3. CycleGAN Loss Balancing

- CycleGAN's final loss includes adversarial losses (for two directions) plus cycle consistency losses. Some implementations also add an identity loss. How do hyperparameters (e.g., λ for cycle consistency) affect the visual quality and structural fidelity of the generated images?

3. Diffusion Model Details

1. Noise Schedule

- Discuss how the beta schedule (linear, cosine, etc.) impacts the forward diffusion process.
- What considerations are there for choosing the noise schedule specifically for medical images?

2. Conditioning & Cross-Attention

- In the code, we see an example of a UNet with cross-attention (e.g., conditioning on T1 to generate T2). How does cross-attention differ from simply concatenating the T1 and noisy T2 slices? Under what scenario does cross-attention lead to a measurable improvement?

3. Inference Speed & Memory

- Diffusion models often require multiple reverse sampling steps. Propose one potential strategy for faster inference in diffusion-based image translation, noting any trade-offs in image quality.

4. Evaluation & Metrics in Medical Image Translation

1. Perceptual vs. Structural Metrics

- Medical imaging tasks require preserving anatomical details. Contrast simpler metrics (MSE, SSIM, PSNR) with perceptual metrics or clinical metrics.
 - What are some pitfalls of relying solely on SSIM or PSNR for evaluating medical images?
2. Clinical Validation
- Beyond numerical metrics, how might one incorporate clinician feedback or domain-expert annotation for validation?

Part B: Implementation Tasks

Task 1: CycleGAN for T1 ↔ T2 MRI Translation

Objective: Implement a CycleGAN to translate between T1 and T2 MRI images (similar to the provided class code).

Steps

1. Data Loading & Preprocessing

- Use the same IXI MRI dataset.
- Organize T1 and T2 scans. (Paired data is helpful but not strictly required; if unpaired, follow standard CycleGAN unpaired approach.)

2. Architectural Modifications

- Modify the default generator to include more residual blocks (e.g., 9 instead of 6).
- Optionally, experiment with a U-Net-style generator.
- Keep the PatchGAN discriminator.

3. Training

- Train CycleGAN for 20 epochs.
- Log losses for each epoch (Generator loss, Discriminator loss, Cycle-consistency loss, etc.).

4. Evaluation & Visualization

- Generate T2 images from T1 inputs (and vice versa).

- Measure PSNR, SSIM.
- Provide side-by-side images (real T1, generated T2, real T2, etc.) in a notebook.

Deliverables

- A Jupyter notebook (CycleGAN_MRI_Translation.ipynb) containing:
 - Code for data loading, model definition, training loop, and evaluation.
 - Graphs of training losses vs. epochs.
 - Visualization of real vs. generated MRI slices.

Task 2: Diffusion Model for T1 → T2 Translation

Objective: Adapt the provided DDPM (or similar diffusion code) to do single-direction T1 → T2 generation. You do *not* need to do both directions.

Steps

1. Dataset & Conditioning

- Load the same MRI dataset but focus on T1 → T2.
- In your diffusion code, treat T2 as the “clean target” you want to sample, and T1 as the “conditioning” or “context.”

2. Model Modifications

- You will be training the model without self-attention and cross-attention because that would be very slow. However, you need to compare the number of trainable model parameters with and without self-attention and cross-attention.

3. Training Procedure

- You can reduce from time-steps from 1,000 to 300–500 timesteps for quick testing if needed.
- Track training loss curves (MSE loss between predicted noise and actual noise).

4. Sampling & Visualization

- Generate T2 slices from T1 slices at inference time, using the reverse diffusion process.

- Show side-by-side results: T1 input, sampled T2, real T2.
- Measure PSNR, SSIM.

Deliverables

- A Jupyter notebook (Diffusion_T1_to_T2.ipynb) with:
 - Updated diffusion code supporting $T1 \rightarrow T2$.
 - Sampled outputs at different reverse diffusion steps (e.g., after 10, 100, 300 steps).
 - Metrics (SSIM, PSNR) on test slices.

Submission Guidelines

Submit a zipped folder named [Lastname]_[Firstname]_HW_03.zip containing:

1. Code Notebooks

- CycleGAN_MRI_Translation.ipynb
- Diffusion_T1_to_T2.ipynb

2. Short Report (PDF or Markdown)

- Written answers for Part A: Conceptual Questions

3. Plots & Visualizations

- Include any training-loss plots, visual comparisons, etc.